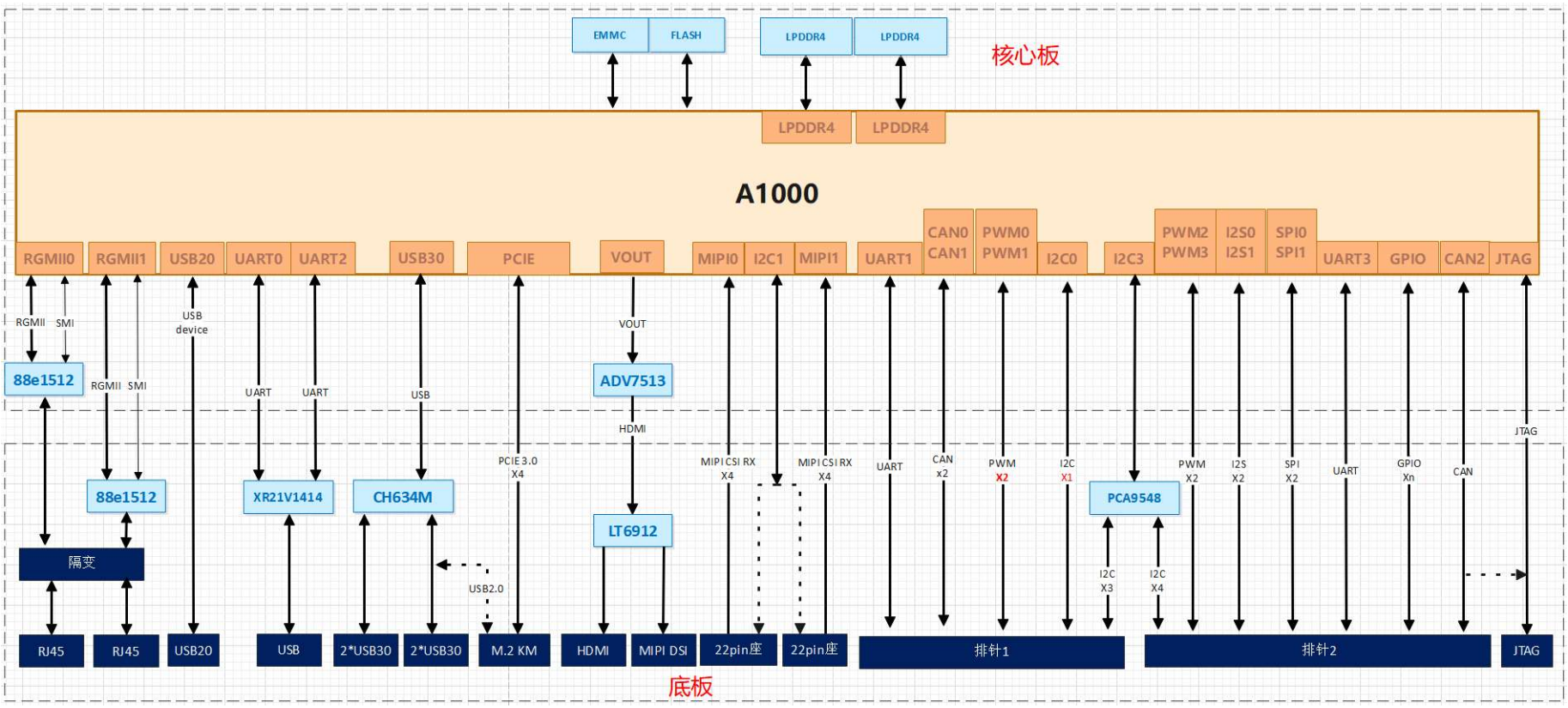
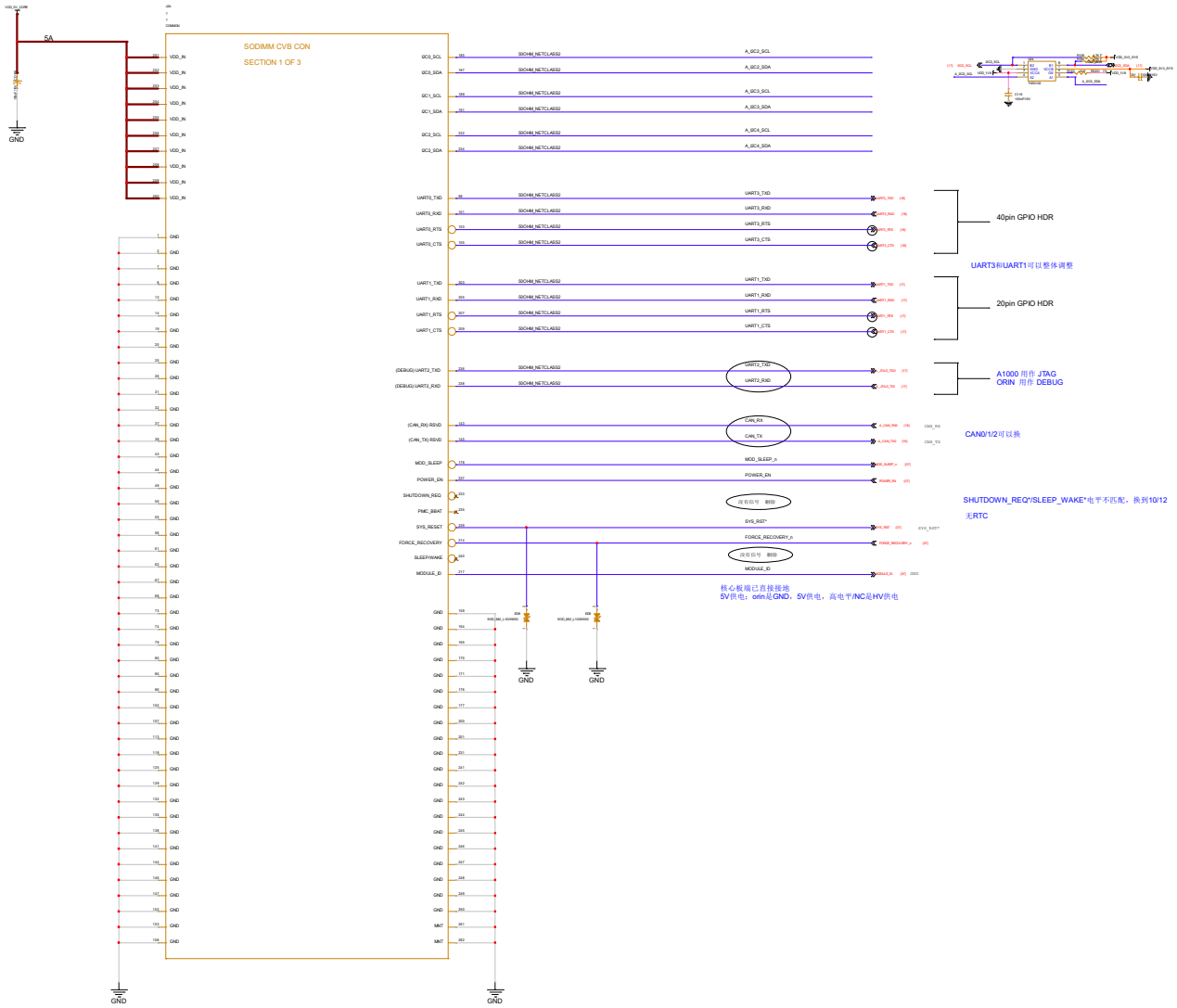
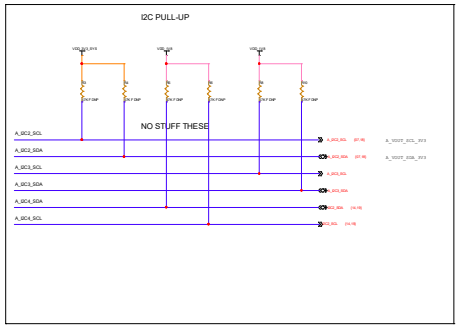


FuZhou X-Speed Information Technology Co.,Ltd

Title: 01.Block Diagram			
Size: A4	Product Model: RC01_CARRIER_V100-20260202F1C	DESIGNER:	Rev: V001
Date: Thursday, December 04, 2025	Sheet 01	of 22	

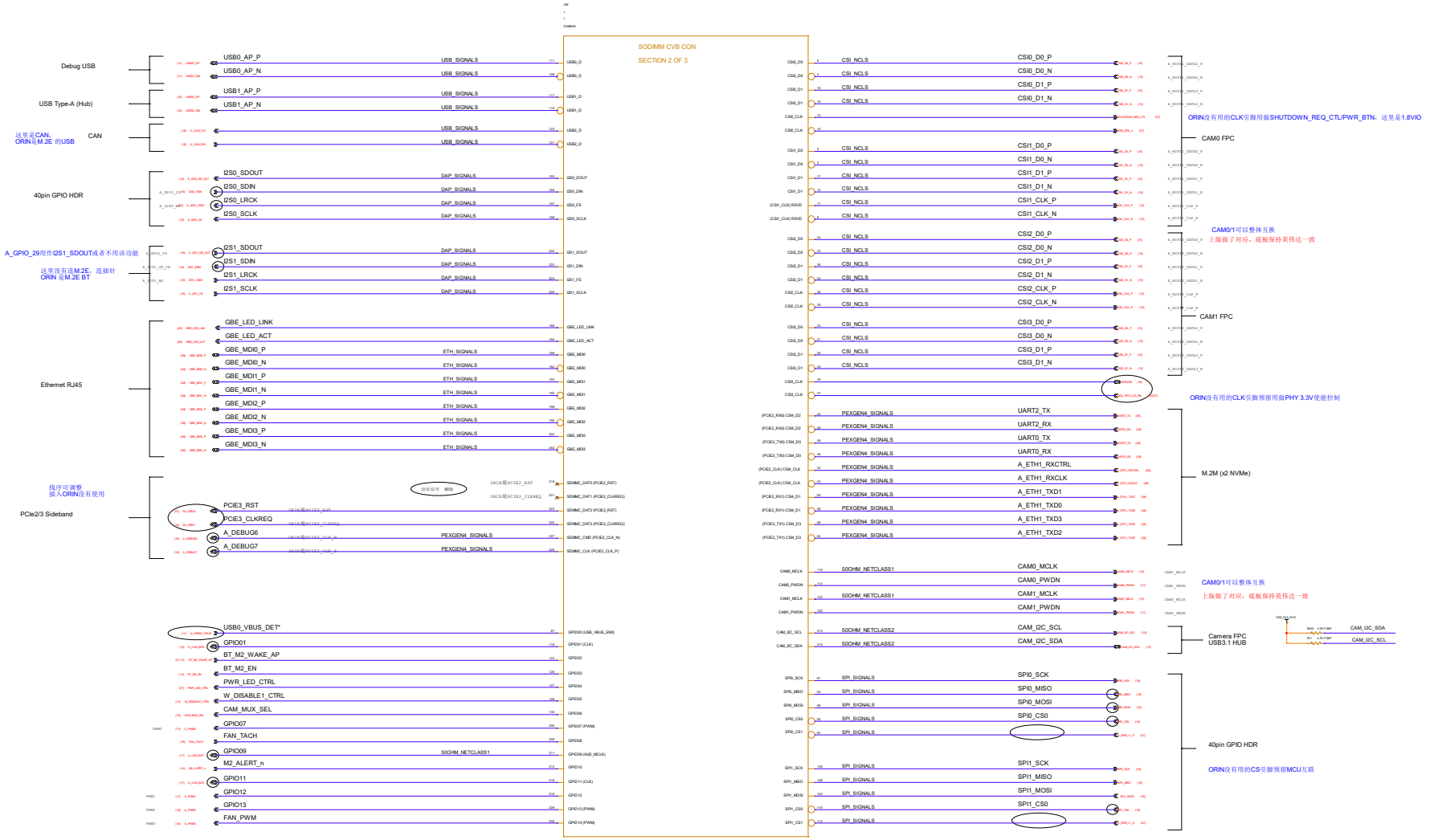
UART, I2C, CAN and GPIOs





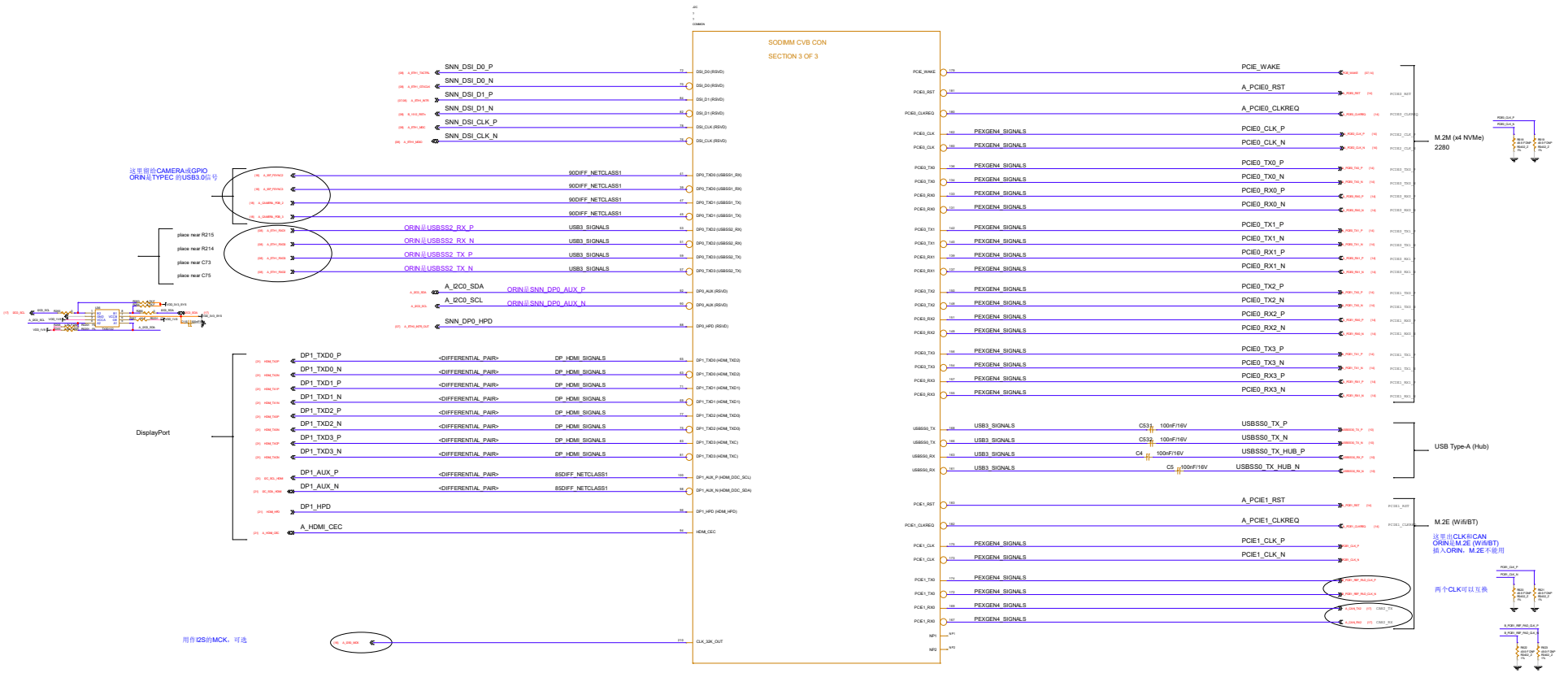
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 02.SODIMM Connector 1_3			
Size: Custom	Product Model: RC01_CARRIER_V100-20250218	DESIGNER: IRC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 02 of 22		

GPIOs, USB2.0, I2S, SPI, GBe and CSI



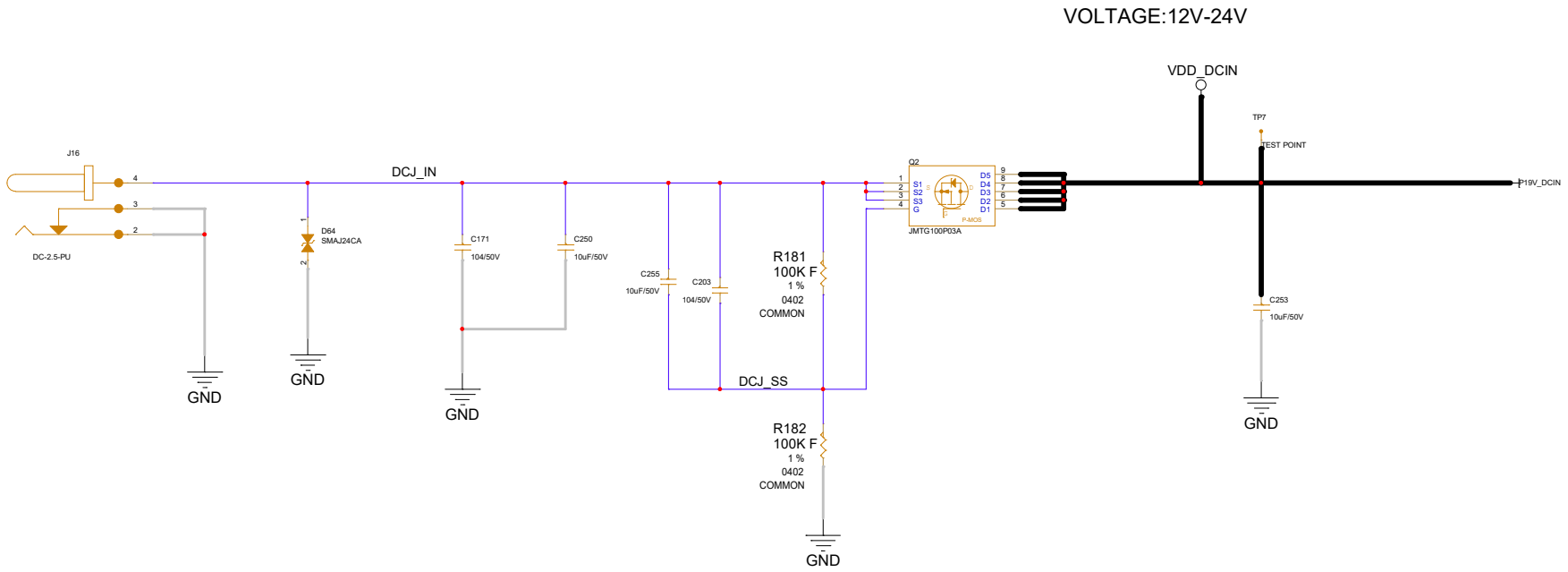
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 03.SODIMM Connector 2_3			
Size: Custom	Product Model: RC01_CARRIER_V100-20250MRC	DESIGNER	Rev: V001
Date: Friday, December 05, 2025	Sheet 03	of 22	

DP, USBSS and PEX

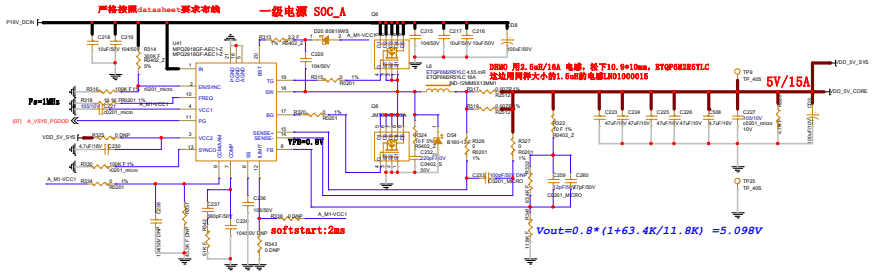


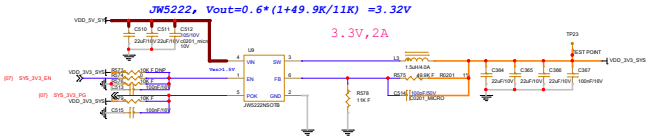
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 04.SODIMM Connector 3_3			
Size: Custom	Product Model: RC01_CARRIER_V100-20250	DESIGNER: SOMRC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 04 of 22		

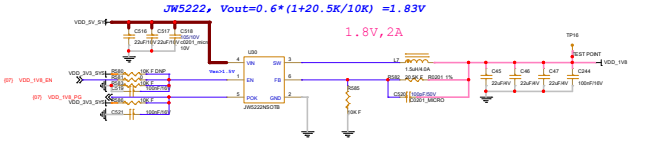
DC JACK POWER PATH

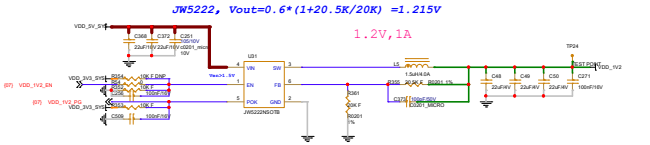


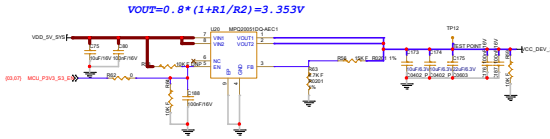
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 05.DC JACK			
Size: A2	Product Model: RC01_CARRIER_V100-2020	DESIGNER: 902F1C	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 05 of 22		

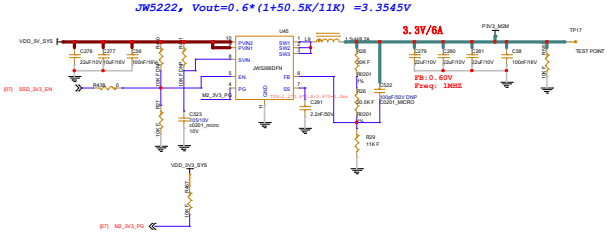


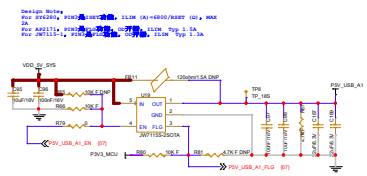


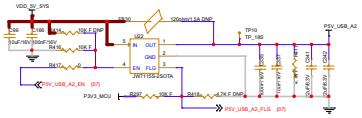


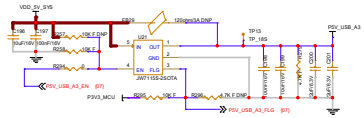


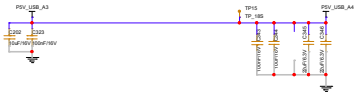








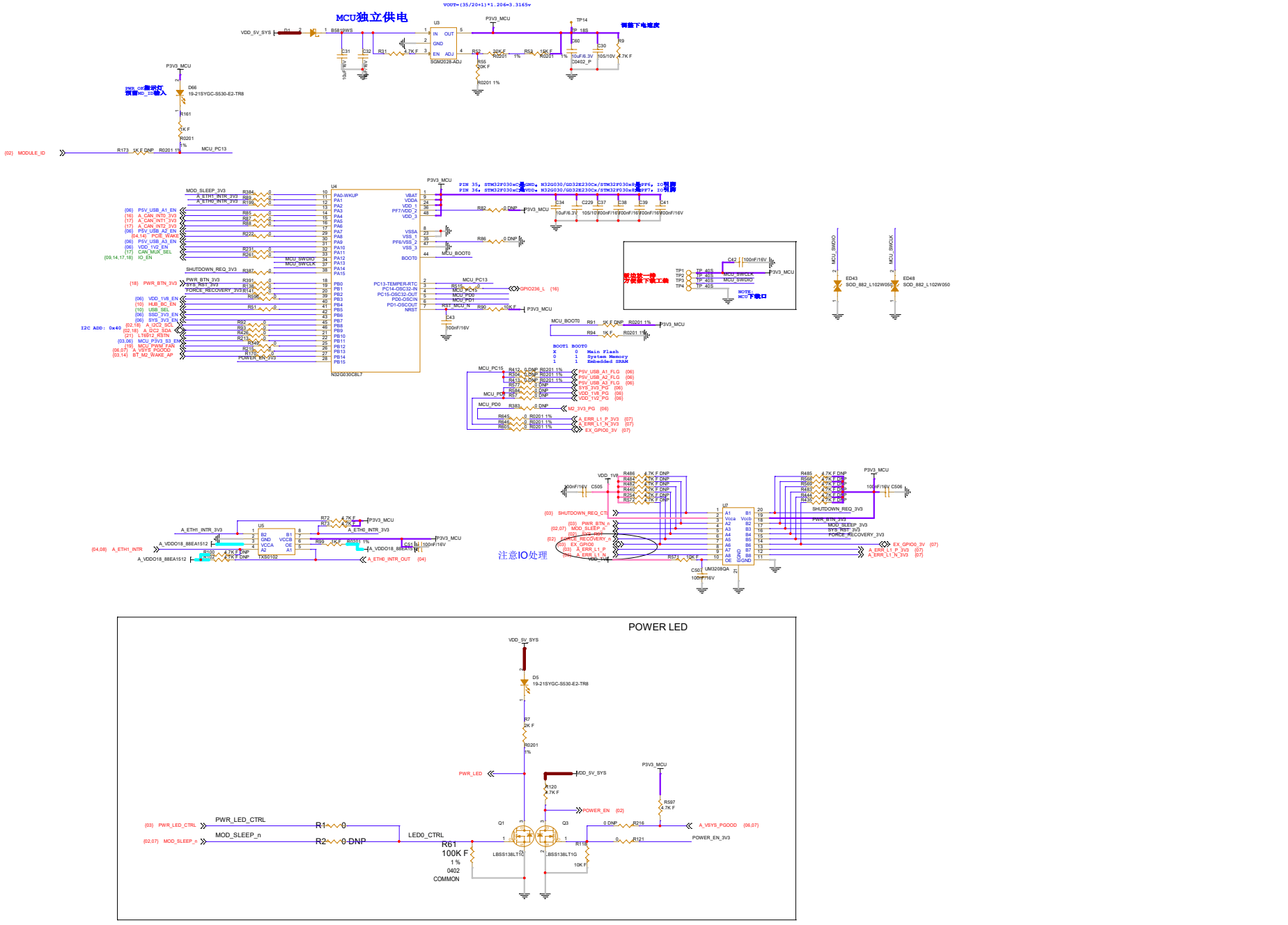




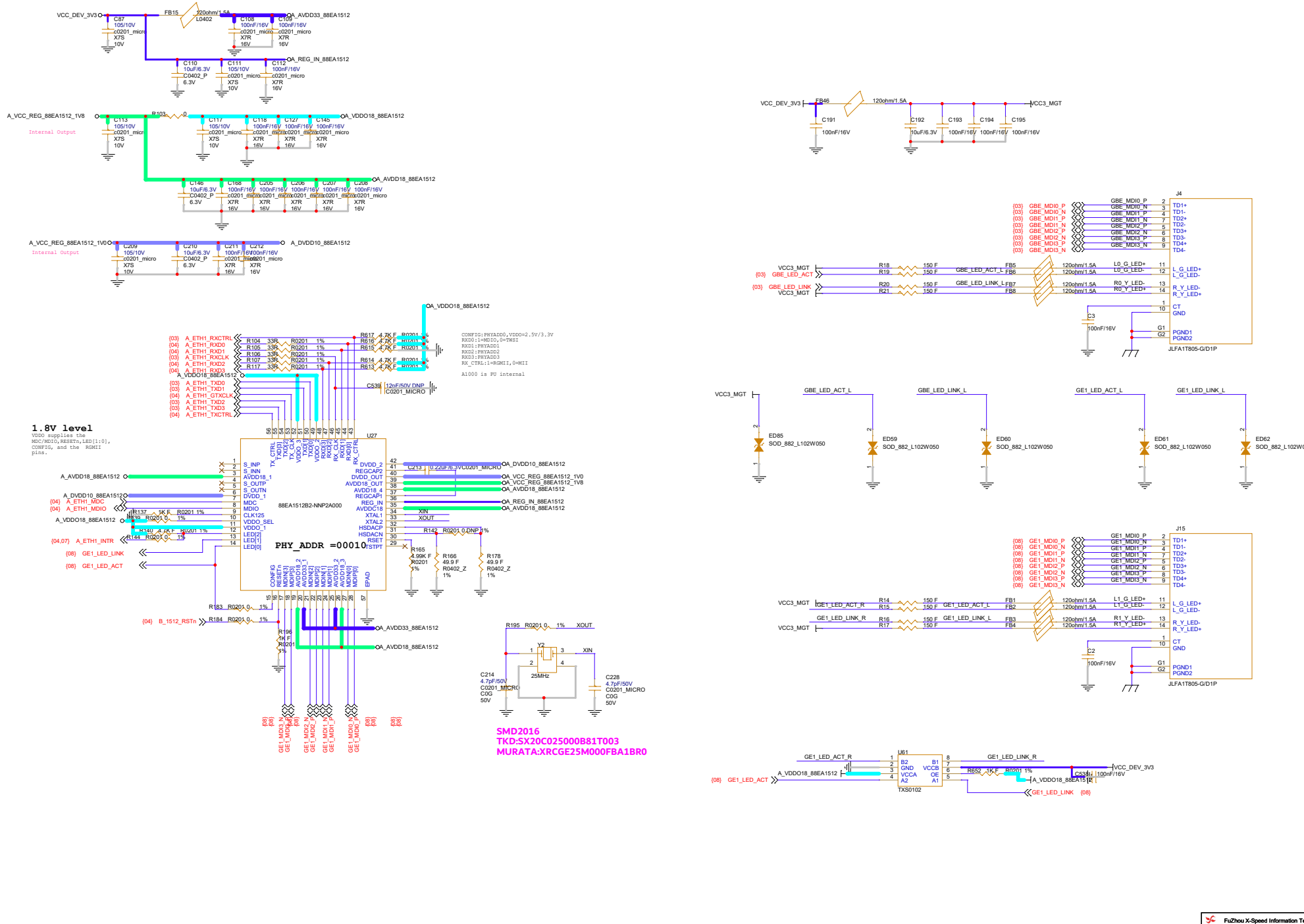


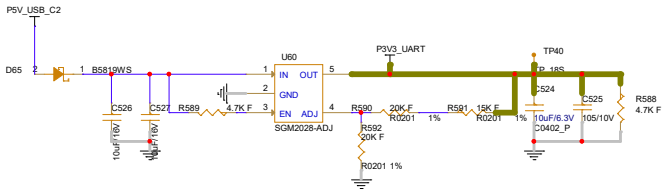


Fuzhou X-Speed Information Technology Co.,Ltd			
Title: 06:PWR			
Size: Custom	Project Model: RDR_CARRIER_V10-3S-2GBHC	Designed:	Rev: V0.1
Date: Wednesday, February 12, 2025	Sheet: 6/8		

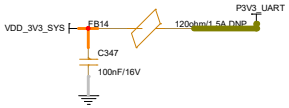


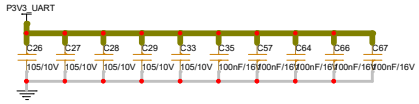
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 07.MCU			
Size: Custom	Product Model: RC01_CARRIER_V100-20	DESIGNER: [illegible]PC	Rev: V001
Date: Wednesday, February 11, 20	Sheet: 07	of 22	

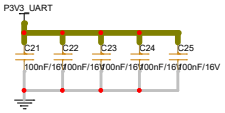

VCC_DEV_3V3
FB15
120ohm/1.5A
L0402
A_AVDD33_88EA1512
A_REG_IN_88EA1512
A_VCC_REG_88EA1512_1V8
Internal Output
A_AVDD18_88EA1512
A_VCC_REG_88EA1512_1V0
A_DVDD10_88EA1512
A_VDDO18_88EA1512
1.8V level
VDDO supplies the MDC/MDIO, RESETn, LED[1:0], CONFIG, and the RGMII pins.
88EA1512B2-NNP2A000
PHY_ADDR =00010
U27
CONFIG:PHYADD0,VDDO=2.5V/3.3V
RXD0:1=MDIO,0=TWSI
RXD1:PHYADD1
RXD2:PHYADD2
RXD3:PHYADD3
RX_CTRL:1=RGMII,0=MII
A1000 is PU internal
A_ETH1_MDC
A_ETH1_MDIO
A_ETH1_INTR
GE1_LED_LINK
GE1_LED_ACT
B_1512_RSTn
XIN
XOUT
25MHz
SMD2016
TKD:SX20C025000B81T003
MURATA:XRCGE25M000FBA1BR0
J4
J15
JLFA1T805-G/D1P
GBE_LED_ACT_L
GBE_LED_LINK_L
GE1_LED_ACT_L
GE1_LED_LINK_L
ED85
ED59
ED60
ED61
ED62
SOD_882_L102W050
U61
TXS0102
FuZhou X-Speed Information Technology Co.,Ltd
08.ETH
Product Model
RC01_CARRIER_V100-20200
DESIGNER
V001
Wednesday, February 11, 2026
Sheet 08 of 22

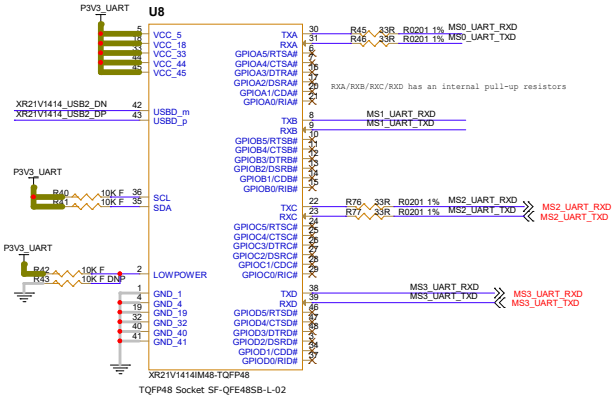


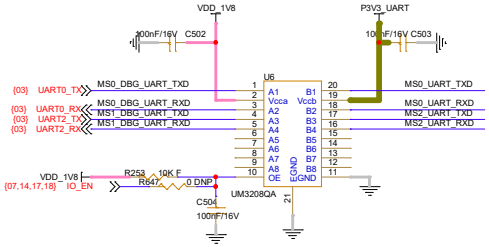
预留系统供电

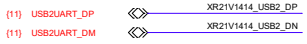












Layout Resistors as 3-PAD

FuZhou X-Speed Information Technology Co.,Ltd			
Title: 09.USB_TO_UART			
Size: A2	Product Model: RC01_CARRIER_V100-20200	DESIGNER: FIC	Rev: V001
Date: Wednesday, February 11, 202	Sheet 09	of 22	

GL3510需要上DR电阻
CH634 需要DNP

HUB_AVDD R432 0 DNP HUB_AVDD2

HUB_3V3 R278 0 DNP HUB_3V3_1

GL3510 SUSP/CFG2引脚悬空
CH634 SUSP/CFG2指示BC使能:
内置上拉电阻，高电平为使能 BC 充电，
悬空或外加下拉电阻置低电平为禁止 BC 充电.

HUB_3V3

BC Enable

{07} HUB_BC_EN — R239 15K F — R2201 1% — SUSP/CFG2

R238 10K F

BC Disable

GL3510 IT03000050
CH634 IT03000076

USB信号中的DP或DN是相对于MOSFET而言

U38

{13} HUB_DS3_TXN	1	TXN_DS3	
{13} HUB_DS3_TXP	2	TXP_DS3	
HUB_AV1V2	3	VP12_3	
{13} HUB_DS3_RXN	4	RXN_DS3	
{13} HUB_DS3_RXP	5	RXP_DS3	
{13} HUB_DS3_DP	6	DP3	
{13} HUB_DS3_DN	7	DM3	
HUB_AVDD2	8	AVDD_2	
DS4_DP	9	DP4	
DS4_DN	10	DM4	
{13} HUB_DS4_TXN	11	TXN_DS4	
{13} HUB_DS4_TXP	12	TXP_DS4	
HUB_AV1V2	13	VP12_4	
{13} HUB_DS4_RXN	14	RXN_DS4	
{13} HUB_DS4_RXP	15	RXP_DS4	
R227 0	16	RTERM	
C119 100nF/16V HUB_CHIP_GND	17	FB	
HUB_SW	18	VSSP	
	19	SW	
HUB_PSV	20	5V	
HUB_3V3	21	5V	
HUB_FN_A	22	V33	
HUB_FN_B	23	FN_A	
HUB_RESETJ	24	FN_B	
HUB_VBUS_IN	25	RESETJ	
HUB_3V3_1	26	VBUS	
	27	DVDD33_1	
HUB_DV1V2	28	DVDD12_3	
HUB_SPI_DO	29	P_SPI_DO	
HUB_SPI_CK	30	P_SPI_CK	
HUB_SPI_DI	31	P_SPI_DI	
HUB_SPI_CZ	32	P_SPI_CZ	
SUSP/CFG2		FLED/FN_C	

CH634W6G/GL3510

	64	DM2
	63	DP2
HUB_3V3	62	DVDD33_2
HUB_DV1V2	61	DVDD12_1
	60	RXP_DS2
	59	RXN_DS2
HUB_AV1V2	58	VP12_2
	57	TXP_DS2
	56	TXN_DS2
HUB_DV1V2	55	DVDD12_2
HUB_AVDD	54	AVDD_1
	53	DM1
	52	DP1
	51	RXP_DS1
	50	RXN_DS1
HUB_AV1V2	49	VP12_1
	48	TXP_DS1
	47	TXN_DS1
HUB_AV1V2	46	AVDD12
HUB_X1_IN	45	X1
HUB_X2_OUT	44	X2
HUB_AVDD	43	AVDD_4
	42	RXP_UP
	41	RXN_UP
HUB_AV1V2	40	VP12_5
	39	TXP_UP
	38	TXN_UP
	37	DP0
HUB_AVDD	36	DM0
HUB_PWRENJ	35	AVDD_3
HUB_OVCURJ	34	PWRENJ
	33	OVCURJ

HUB_DS2_DN	{12}	
HUB_DS2_DP	{12}	
HUB_DS2_RXP	{12}	
HUB_DS2_RXN	{12}	
HUB_DS2_TXP	{12}	
HUB_DS2_TXN	{12}	
HUB_DS1_DN	{12}	
HUB_DS1_DP	{12}	
HUB_DS1_RXP	{12}	
HUB_DS1_RXN	{12}	
HUB_DS1_TXP	{12}	
HUB_DS1_TXN	{12}	
USBSS30_TX_P	{04}	
USBSS30_TX_N	{04}	
USBSS30_RX_P	{04}	
USBSS30_RX_N	{04}	
USB30_DP	{03}	
USB30_DM	{03}	

PIN16 :
GL3510使用20k接GND;
CH634使用0R接地

PIN20: 5V Power input. If using external regulator, this pin should be connected with 3.3V
PIN21: 5V-to-3.3V regulator Vout & 3.3 input

内置集成了5V-to-1.2V 120 和 5V-to-1.2V Buck
可以使用5V供电（最高功耗1.2W），或使用外部3.3V(300mA)和1.2V(600mA)供电

GL3510使用25MHz晶体
CH634 使用24MHz晶体

C177 12pF/50V C0201_MICRO
HUB_X2_OUT — R305 0 — R302 1M F DNP — Y3 24MHz — R301 0 — HUB_X1_IN
C178 12pF/50V C0201_MICRO

HUB_SPI_CZ
HUB_SPI_DI
HUB_3V3
R281 10K F
TP4 TEST POINT ×5
R285 10K F
C120 100nF/16V
HUB_SPI_CK
HUB_SPI_DO

用前测试A，用于SPI flash程序更新调试

HUB_PSV R282 10K F HUB_RESETJ
R283 20K
C36 105/10V
R0201 1%

VDD_5V_SYS — FB43 120ohm/1.5A — HUB_P5V
C44 22uF/10V 100nF/16V
C131 100nF/16V
C136 100nF/16V
C137 22uF/10V
HUB_CHIP_GND R289 0

HUB_P5V R284 4.7K F — HUB_VBUS_IN R290 10K F
HUB_3V3 R285 10K F DNP — HUB_PWRENJ R291 10K F DNP
HUB_3V3 R286 10K F — HUB_OVCURJ

PWRENJ:
Pull high to support low-active power switch
Floating to support poly-fuse
Pull down to support high-active power switch

HUB_3V3 R287 10K F — HUB_FN_A
HUB_3V3 R288 10K F — HUB_FN_B R292 10K F DNP

FN_A Pull High: NO Strapping
FN_A Floating : Set Port3 as Non_Removable

FN_B Pull High: NO Strapping
FN_B Pull Low : Port Disabled for Port4
FN_B Floating : Set Port4 as Non_Removable

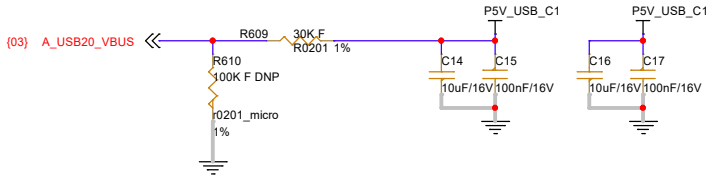
Internal 1.2V Power Circuit

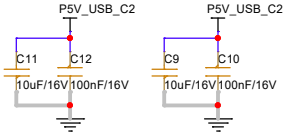
HUB_SW — L4 2.2uH/1.6A — HUB_FB
R293 2.2 F
R0402_Z 1%
C172 102/50V
C153 10uF/6.3V
HUB_DV1V2 — 120ohm/1.5A — FB13 — HUB_AV1V2
C151 10uF/6.3V
C152 10uF/6.3V
C138 C139 C140 C141 C144 C147 100nF/16V
HUB_DV1V2
C148 C149 C150 100nF/16V
HUB_3V3 — FB12 600ohm/1.5A — HUB_AVDD
C160 10uF/6.3V
C161 C162 100nF/16V
C154 C157 C158 C159 100nF/16V

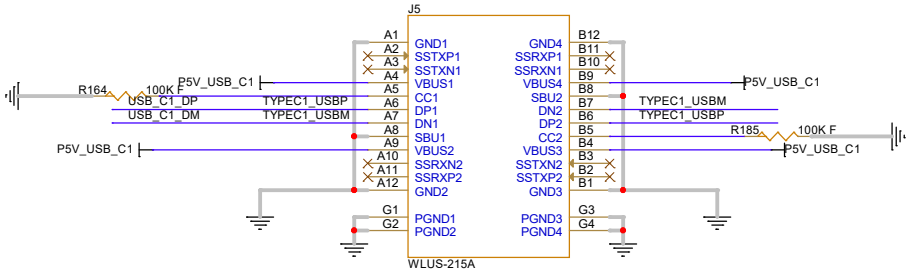
VDD_3V3_SYS
C121 100nF/16V
C122 100nF/16V
DS4_DP
DS4_DN
U2 RS2227
VCC
D-
D+
SEL
GND
HSD1-
HSD1+
HSD2-
HSD2+
OE
{07} USB_SEL — R187 100K F
HD_E_USB2_DN {14}
HD_E_USB2_DP {14}
HUB_DS4_DN {13}
HUB_DS4_DP {13}

FuZhou X-Speed Information Technology Co.,Ltd

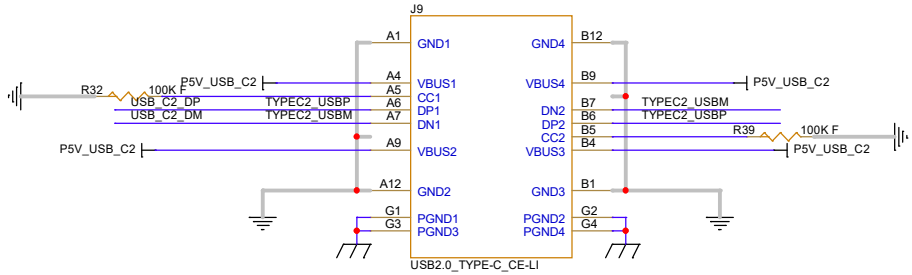
File	10.USB_HUB		
Size	Product Model	DESIGNER	Rev
Custom	RC01_CARRIER_V100-20200	WBFYC	V001
Date:	Wednesday, February 11, 2026	Sheet 10	of 22

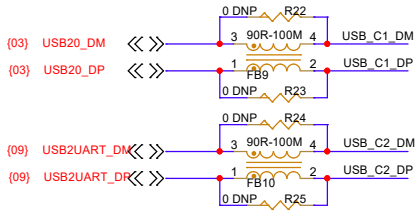




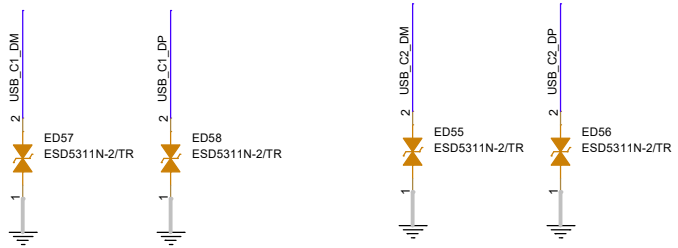


在背面，调试用，PGND，连到螺丝孔，螺丝孔用GND

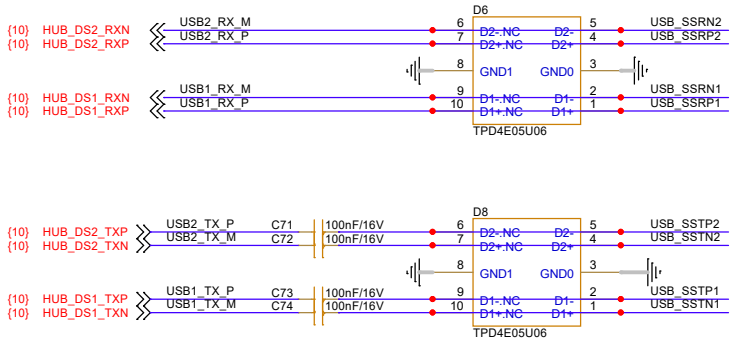


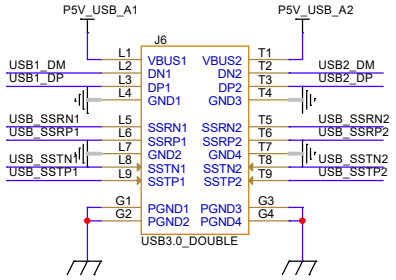


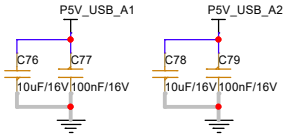
共模电感与电阻PIN量一起，做兼容

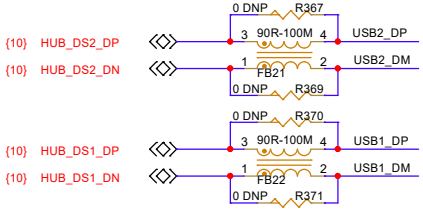


FuZhou X-Speed Information Technology Co.,Ltd			
Title: 11.USB_TYPE_C			
Size: A3	Product Model: RC01_CARRIER_V100-2026020	DESIGNER: LHC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 11 of 22		

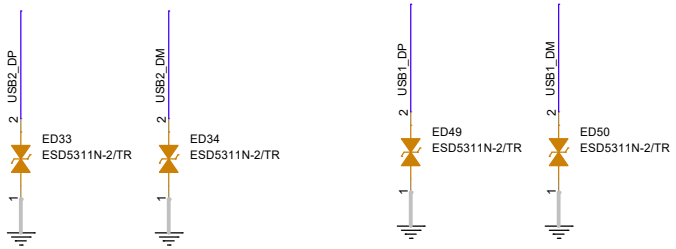




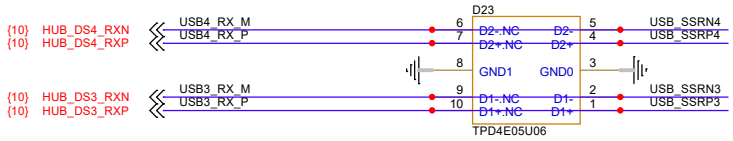


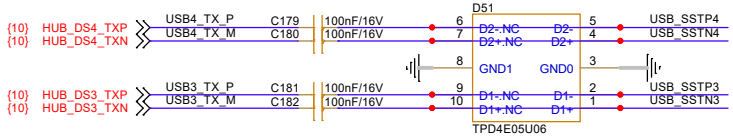


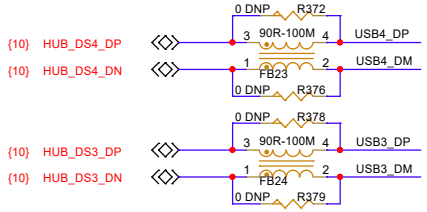
共模电感与电阻PIN叠一起，做兼容



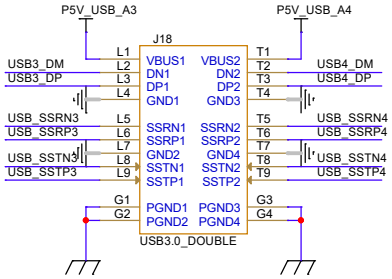
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 12.USB30_A			
Size: A3	Product Model: RC01_CARRIER_V100-2026	DESIGNER: 2026.1 PC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 12 of 22		

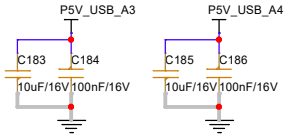


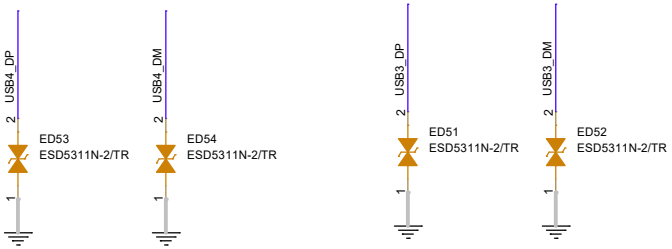




共模电感与电阻PIN叠一起，做兼容

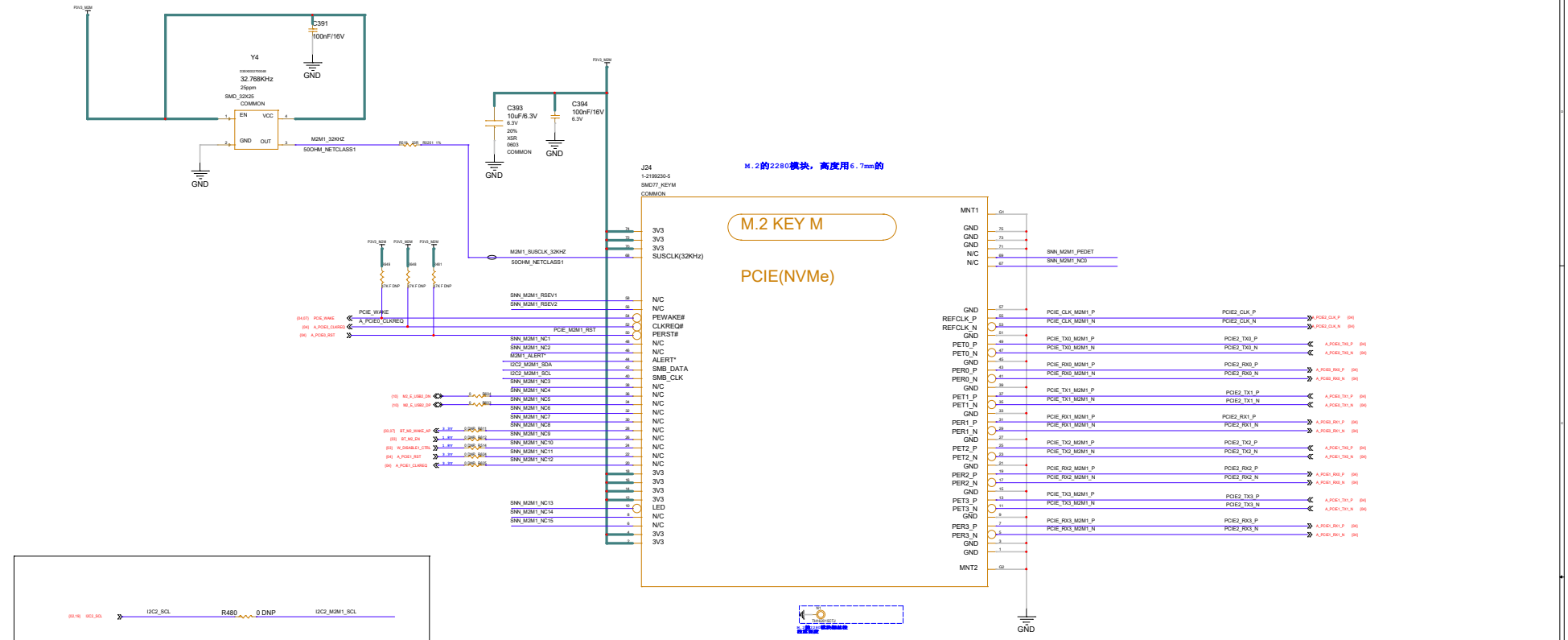


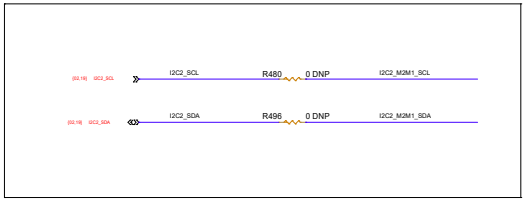


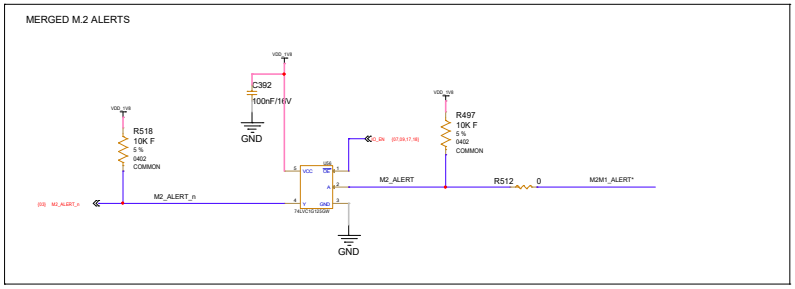


FuZhou X-Speed Information Technology Co.,Ltd			
Title: 13.USB30_B			
Size: A3	Product Model: RC01_CARRIER_V100-2026	DESIGNER: LHC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 13 of 22		

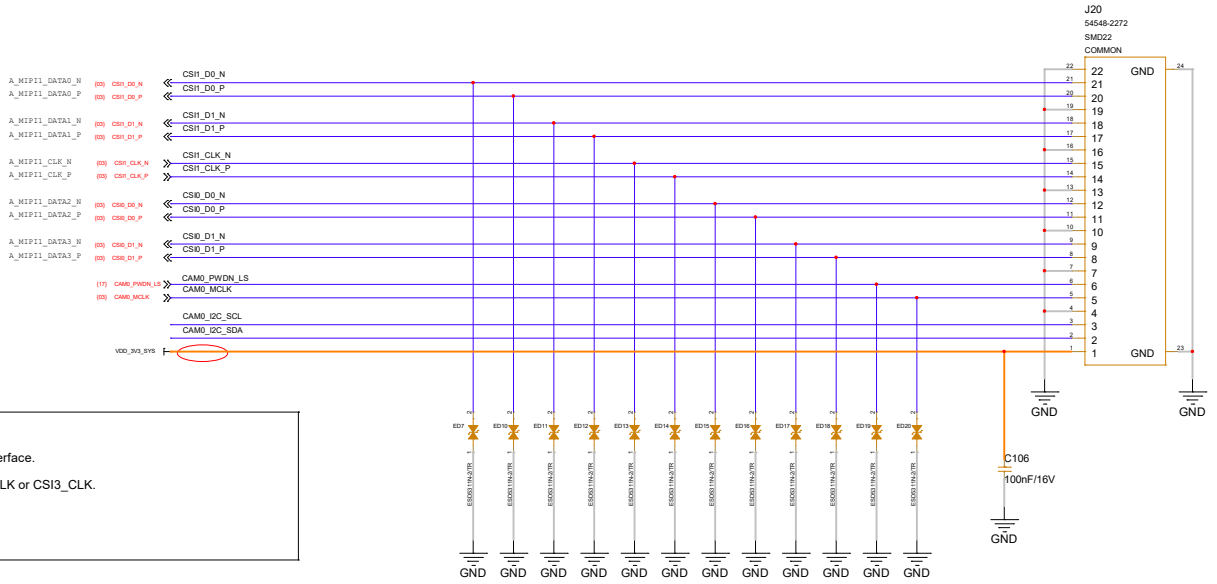
M.2 KEY-M (NVME)


C391
100nF/16V
Y4
32.768KHz
25ppm
SMD_3.2X2.5
COMMON
GND
M2M1_32KHZ
50OHM_NETCLASS1
C393
10uF/6.3V
6.3V
20%
X5R
0603
COMMON
C394
100nF/16V
6.3V
J24
1-2199230-5
SMD77_KEYM
COMMON
M.2的2280模块，高度用6.7mm的
M.2 KEY M
PCIE(NVMe)
3V3
SUSCLK(32KHz)
M2M1_SUSCLK_32KHZ
PCIE_WAKE
A_PCIE7_CLKREQ
PCIE_M2M1_RST
SNN_M2M1_RSEV1
SNN_M2M1_RSEV2
SNN_M2M1_NC1
SNN_M2M1_NC2
SNN_M2M1_NC3
SNN_M2M1_NC4
SNN_M2M1_NC5
SNN_M2M1_NC6
SNN_M2M1_NC7
SNN_M2M1_NC8
SNN_M2M1_NC9
SNN_M2M1_NC10
SNN_M2M1_NC11
SNN_M2M1_NC12
SNN_M2M1_NC13
SNN_M2M1_NC14
SNN_M2M1_NC15
PEWAKE#
CLKREQ#
PERST#
ALERT*
SMB_DATA
SMB_CLK
LED
MNT1
MNT2
SNN_M2M1_PEDET
SNN_M2M1_NC0
PCIE_CLK_M2M1_P
PCIE_CLK_M2M1_N
PCIE_TX0_M2M1_P
PCIE_TX0_M2M1_N
PCIE_RX0_M2M1_P
PCIE_RX0_M2M1_N
PCIE_TX1_M2M1_P
PCIE_TX1_M2M1_N
PCIE_RX1_M2M1_P
PCIE_RX1_M2M1_N
PCIE_TX2_M2M1_P
PCIE_TX2_M2M1_N
PCIE_RX2_M2M1_P
PCIE_RX2_M2M1_N
PCIE_TX3_M2M1_P
PCIE_TX3_M2M1_N
PCIE_RX3_M2M1_P
PCIE_RX3_M2M1_N


I2C2_SCL
R480
0 DNP
I2C2_M2M1_SCL
I2C2_SDA
R496
0 DNP
I2C2_M2M1_SDA


MERGED M.2 ALERTS
C392
100nF/16V
R518
10K F
5 %
0402
COMMON
R497
10K F
5 %
0402
COMMON
M2_ALERT_n
M2_ALERT
R512
0
M2M1_ALERT*
GND

CSI CAM CONNECTORS



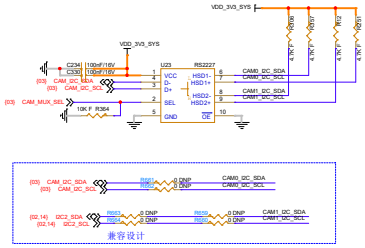
Note:

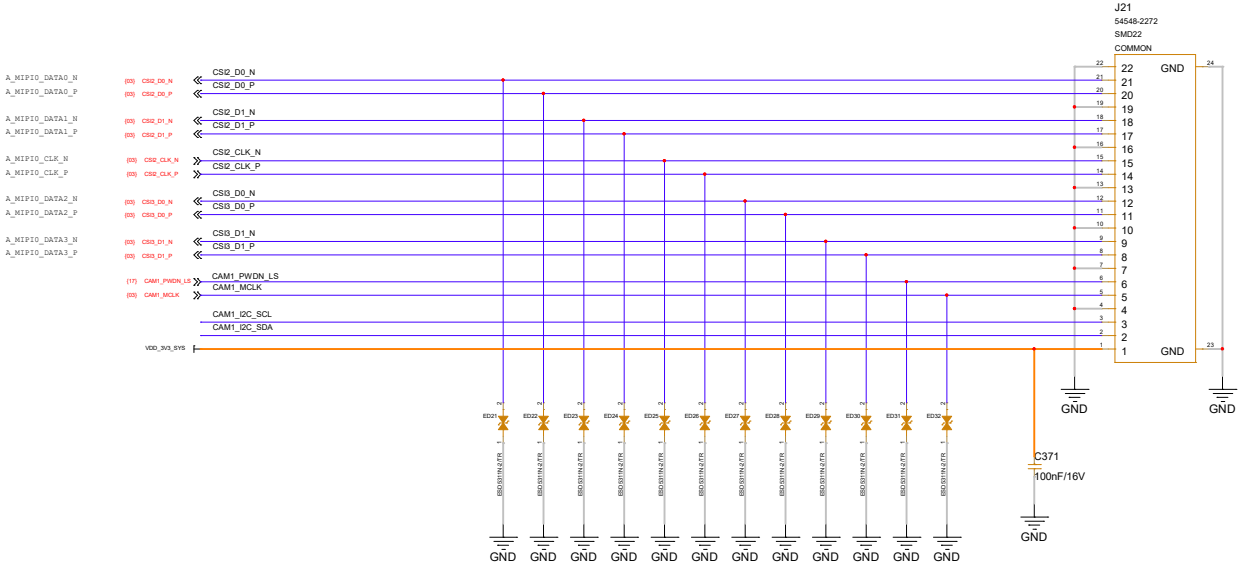
x4 camera support in current software only works with the lower clock lane of the x4 clock interface.

This means that CSI0_CLK or CSI2_CLK are supported for x4 configurations, but not CSI1_CLK or CSI3_CLK.

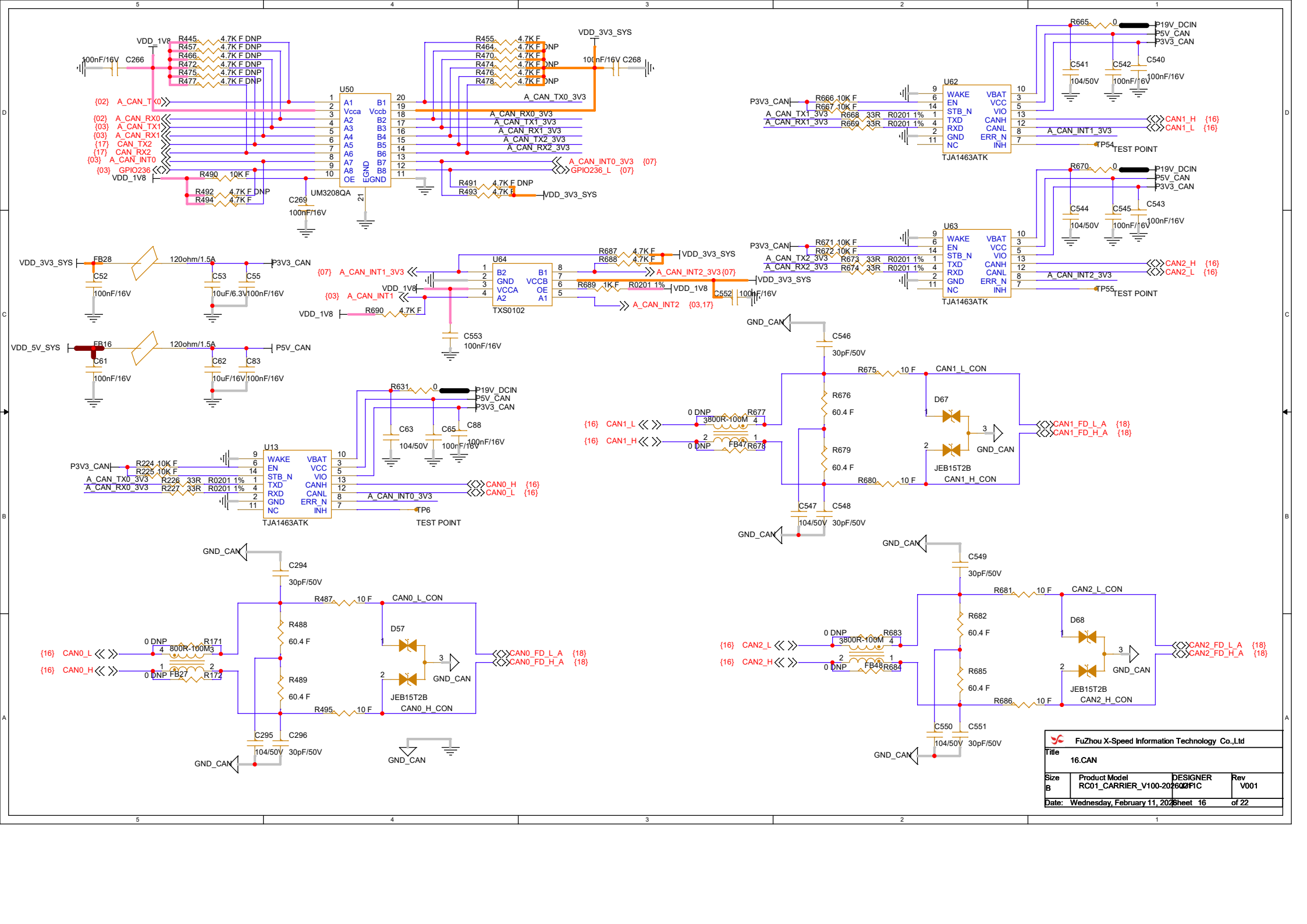
Since J20 has CSI1_CLK, only a x2 camera interface is supported.

J21 which uses CSI2_CLK can support both x2 or x4 camera interfaces.



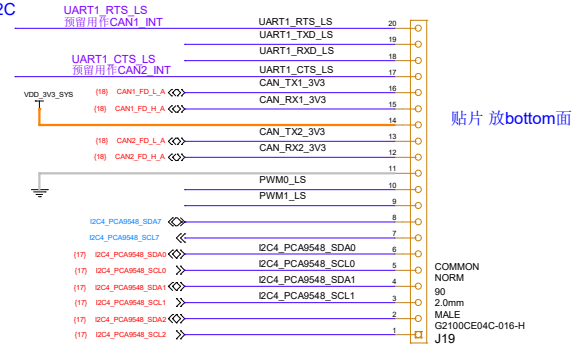


Fuzhou X-Speed Information Technology Co.,Ltd			
Title: 15.CAMERA			
Size: A1	Product Model: RC01_CARRIER_V100-2020	DESIGNER: 33JME	Rev: V001
Date: Wednesday, February 11, 2026	Sheet: 15	of: 22	

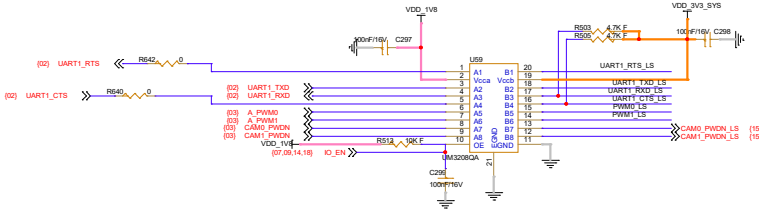

U50
UM3208QA
A_CAN_TX0_3V3
A_CAN_RX0_3V3
A_CAN_TX1_3V3
A_CAN_RX1_3V3
A_CAN_TX2_3V3
A_CAN_RX2_3V3
A_CAN_INT0_3V3
GPIO236_L
U62
TJA1463ATK
CAN1_H
CAN1_L
A_CAN_INT1_3V3
TP54
TEST POINT
U63
CAN2_H
CAN2_L
A_CAN_INT2_3V3
TP55
U64
TXS0102
A_CAN_INT1_3V3
A_CAN_INT2_3V3
A_CAN_INT2
U13
CAN0_H
CAN0_L
A_CAN_INT0_3V3
TP6
P19V_DCIN
P5V_CAN
P3V3_CAN
VDD_3V3_SYS
VDD_1V8
VDD_5V_SYS
GND_CAN
CAN1_L_CON
CAN1_H_CON
CAN2_L_CON
CAN2_H_CON
CAN0_L_CON
CAN0_H_CON
JEB15T2B
CAN1_FD_L_A
CAN1_FD_H_A
CAN2_FD_L_A
CAN2_FD_H_A
CAN0_FD_L_A
CAN0_FD_H_A
FuZhou X-Speed Information Technology Co.,Ltd
16.CAN
Product Model
RC01_CARRIER_V100-20260227
DESIGNER
Rev
V001
Wednesday, February 11, 2026
Sheet 16 of 22

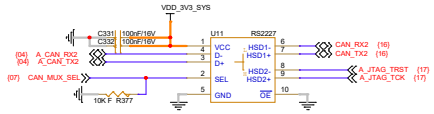
20PIN GPIO EXPANSION HDR

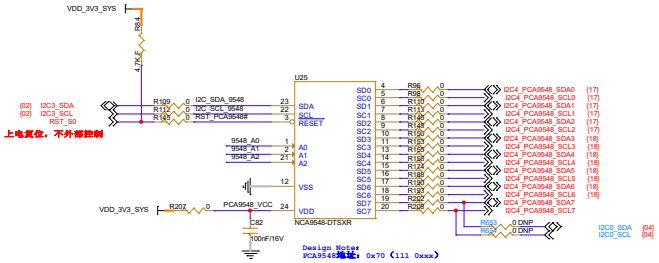
2*CAN-FD(TXRX)
2*PWM
1*UART
4*I2C



贴片 放bottom面

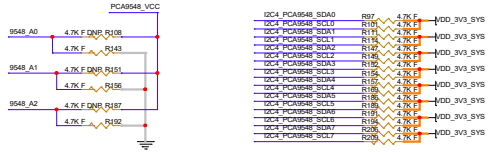


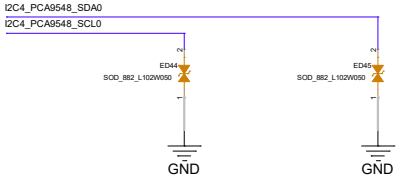


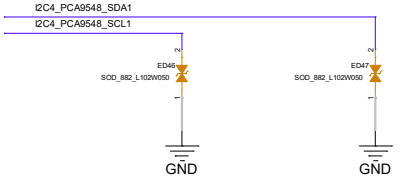


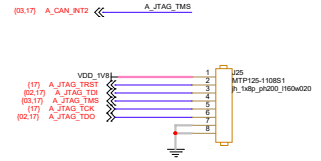
上电复位，不外部控制

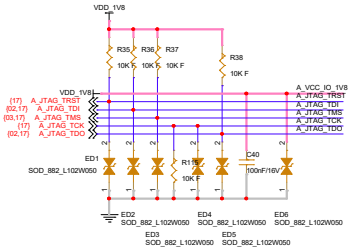
Design Notes
PCA9548地址：0x70（111 0xxx）





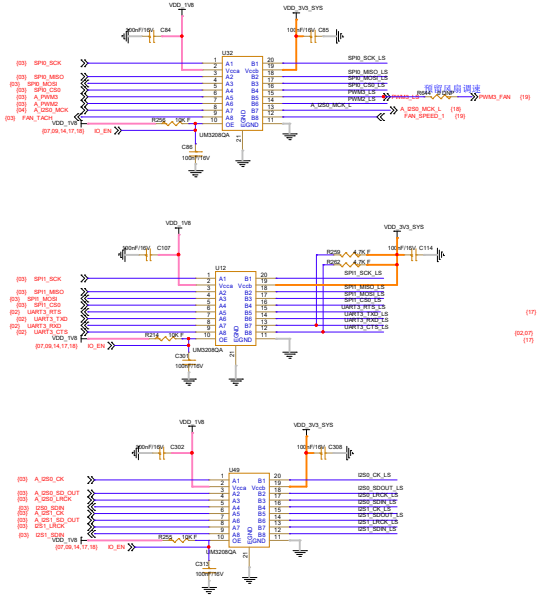


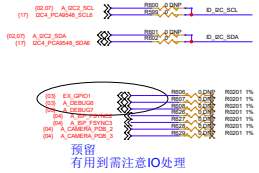


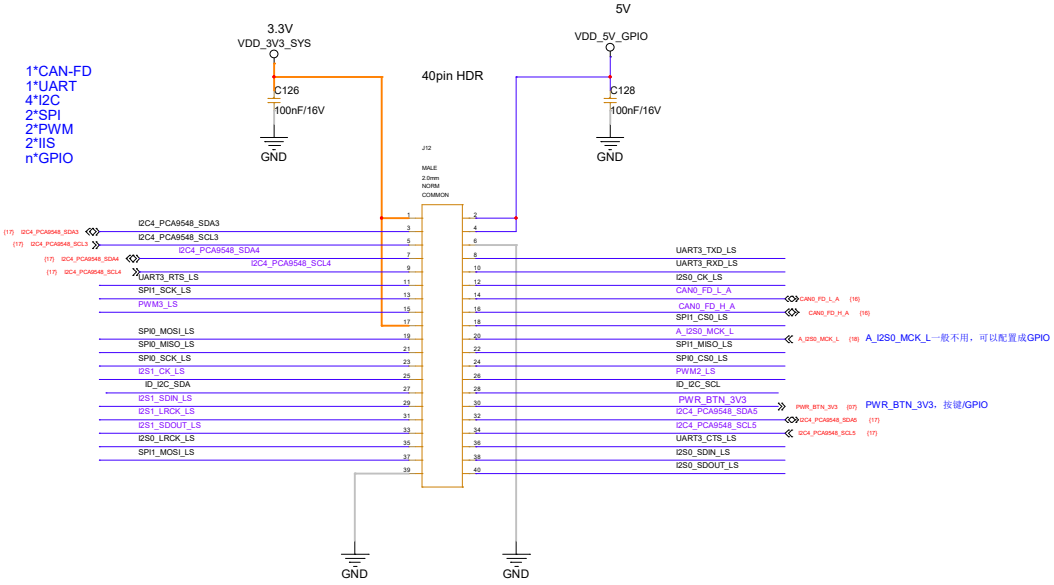


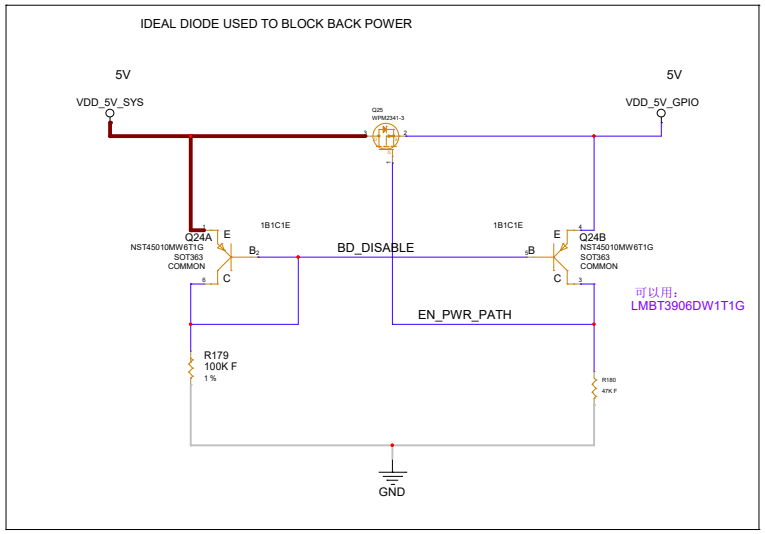
Fuzhou X-Speed Information Technology Co.,Ltd			
Title	17.CONNETOR_20P		
Size: Custom	Product Model: RC01_CARRIER_V100-2020	DESIGNER: 028RC	Rev: V001
Date: Wednesday, February 11, 2020	Sheet 17 of 22		

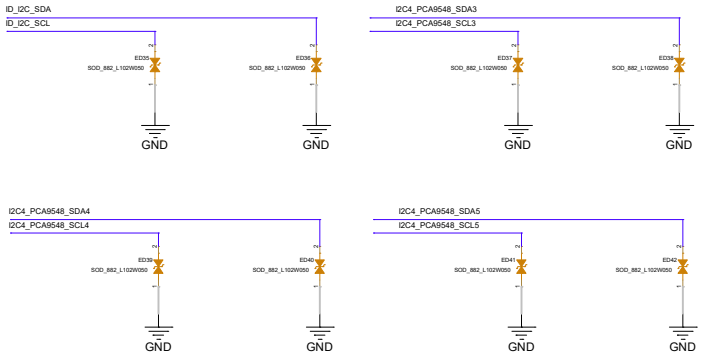
40PIN GPIO EXPANSION HDR





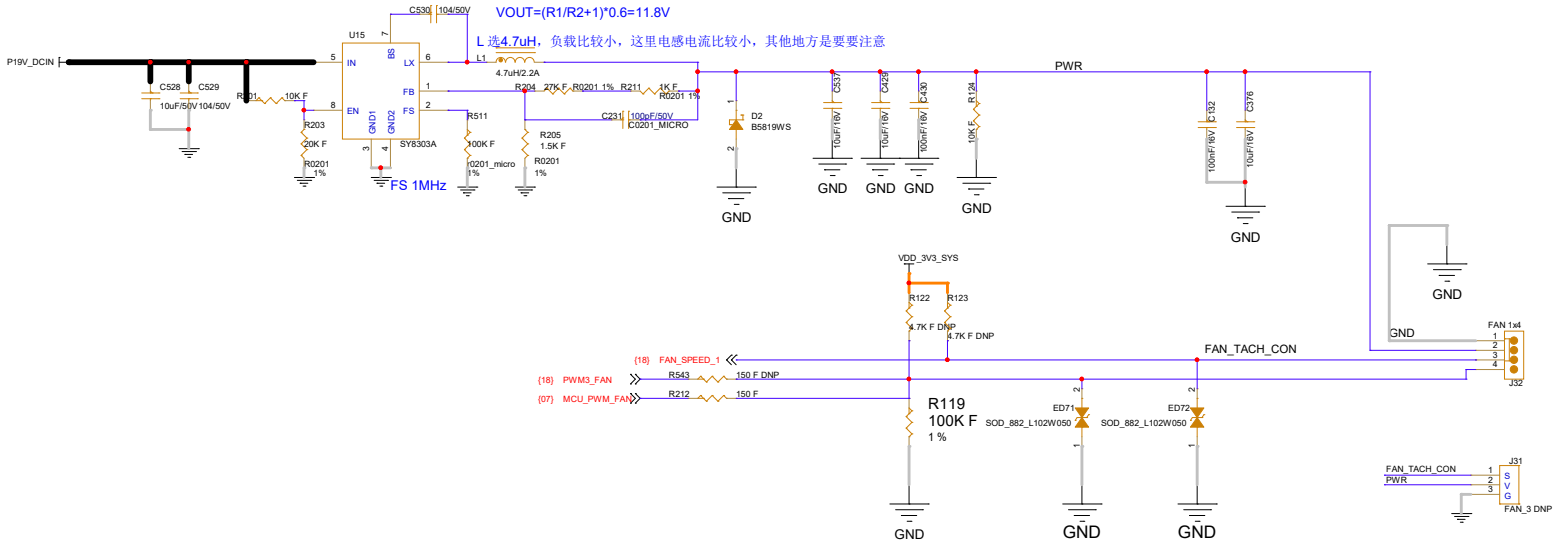




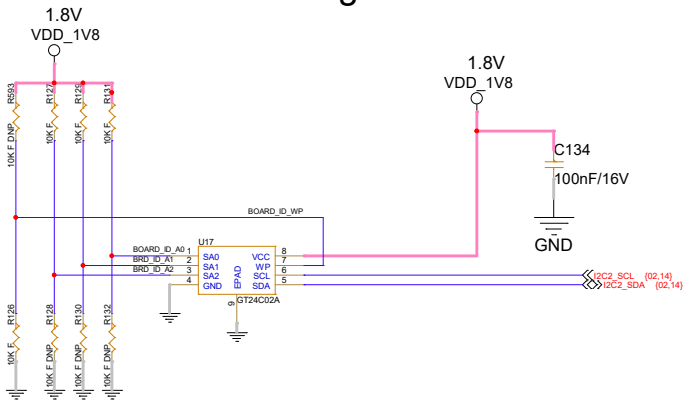


FuZhou X-Speed Information Technology Co.,Ltd			
Title: 18.CONNECTOR_40P			
Size: A1	Product Model: RC01_CARRIER_V100-20200228	DESIGNER:	Rev: V001
Date: Wednesday, February 11, 2026	Sheet: 18	of 22	

SoC Fan Header 4pin



Carrier board config

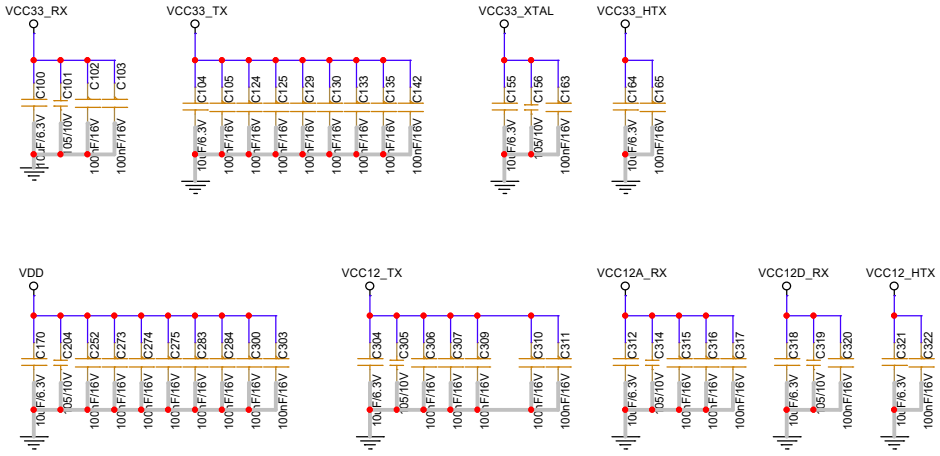


7'b1010xxx

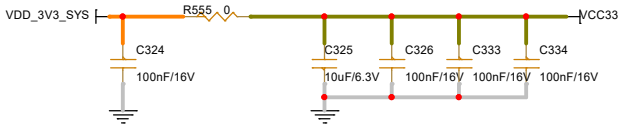
7'b1010111 = 7'h57

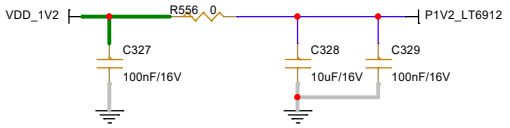
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 19.FAN_EEPROM			
Size: A2	Product Model: RC01_CARRIER_V100-20200	DESIGNER: I2C	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 19 of 22		

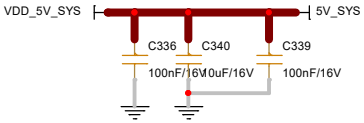
Decouple capacitance: Close IC power pin

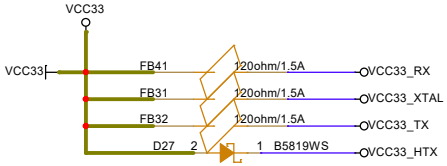


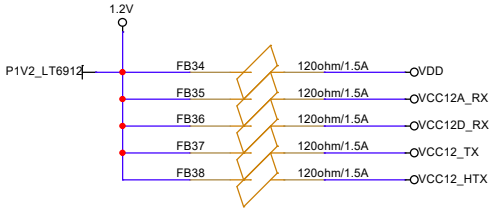
POWER-IN





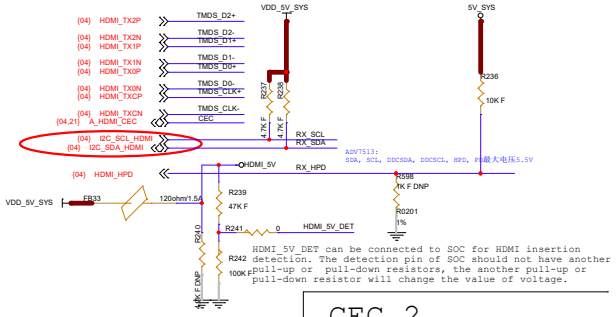






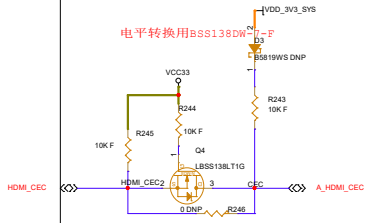
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 20.LT6912_PWR			
Size: A3	Product Model: RC01_CARRIER_V100-2026020	DESIGNER: FPC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 20 of 22		

HDMI INPUT



CEC 2

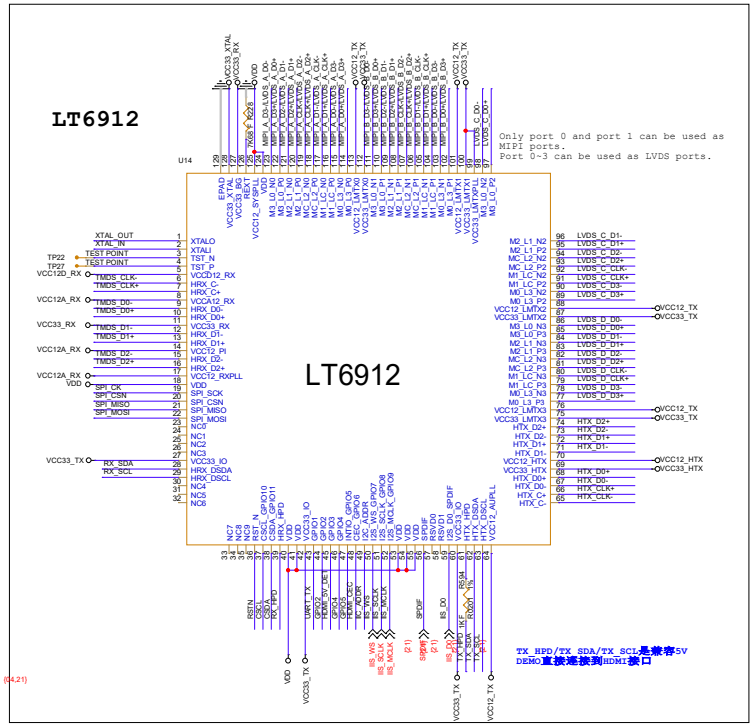
This circuit can't pass the CTS test, but this circuit can be reserved.
This circuit also supports the function of CEC.



CEC 1

This circuit is used to pass CTS test (Test ID 8-14).

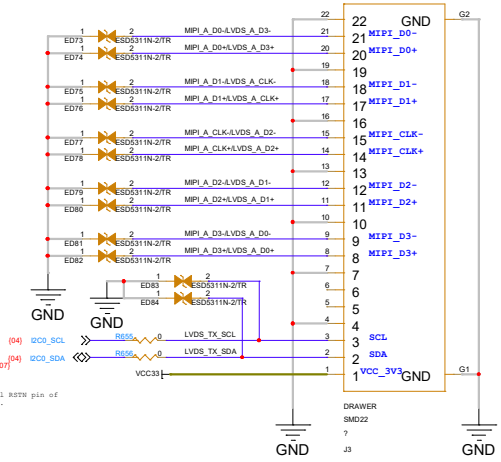
LT6912



MIPI/LVDS_OUT

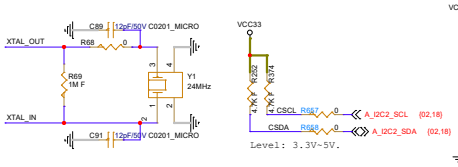
The MIPI and LVDS interface definitions of other ports are the same as port 0.
Only port 0 and port 1 can be used as MIPI.Port 0~3 can be used as LVDS.

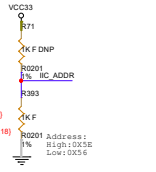
MIPI/LVDS	MIPI	LVDS
MIPI_A_D3-/LVDS_A_D0-	MIPI_lane_data3-	LVDS_lane_data0-
MIPI_A_D3+/LVDS_A_D0+	MIPI_lane_data3+	LVDS_lane_data0+
MIPI_A_D2-/LVDS_A_D1-	MIPI_lane_data2-	LVDS_lane_data1-
MIPI_A_D2+/LVDS_A_D1+	MIPI_lane_data2+	LVDS_lane_data1+
MIPI_A_CLK-/LVDS_A_D2-	MIPI_lane_clock-	LVDS_lane_data2-
MIPI_A_CLK+/LVDS_A_D2+	MIPI_lane_clock+	LVDS_lane_data2+
MIPI_A_D1-/LVDS_A_CLK-	MIPI_lane_data1-	LVDS_lane_clock-
MIPI_A_D1+/LVDS_A_CLK+	MIPI_lane_data1+	LVDS_lane_clock+
MIPI_A_D0-/LVDS_A_D3-	MIPI_lane_data0-	LVDS_lane_data3-
MIPI_A_D0+/LVDS_A_D3+	MIPI_lane_data0+	LVDS_lane_data3+

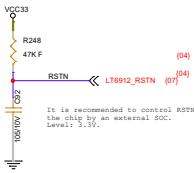


抽屉式下触点

跟树莓派的结序反着，这样实物才是对的

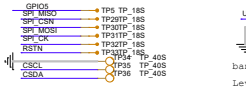


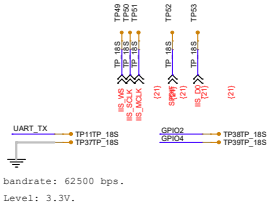




Test Points

SPI test points can be deleted



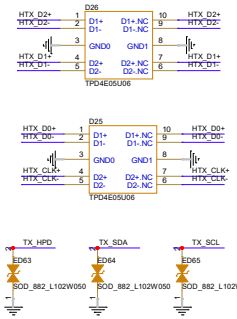


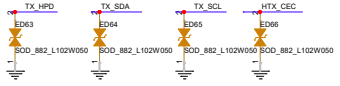
bandrate: 62500 bps.
Level: 3.3V.

CSDA/CSCL:
I2C interface can be used to upgrade the firmware.
You can upgrade the firmware by "USB to I2C" adapter or the I2C interface of an external SOC.

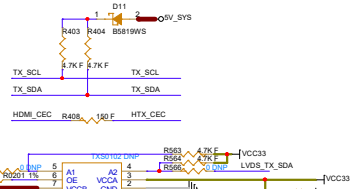
GPIO5:Interrupt I/O.Need to be connected to soc when soc want to read the edid of sink.

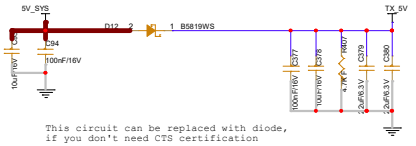
Don't delete the 1M ohms resistor.



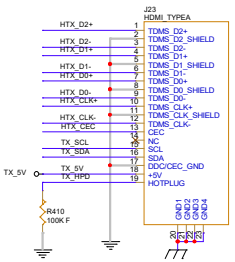


ESD
It can be deleted, if you don't need.





This circuit can be replaced with diode, if you don't need CTS certification



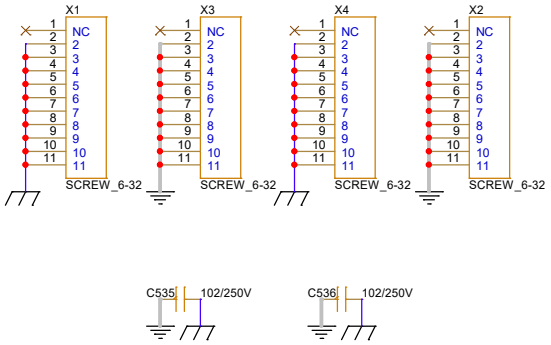
FuZhou X-Speed Information Technology Co.,Ltd			
Title: 21.LT6912			
Size: Custom	Product Model: RC01_CARRIER_V100-2022	DESIGNER: 2022/FC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 21 of 22		

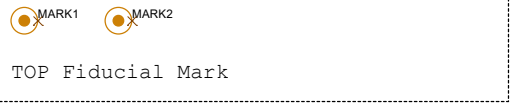
N2
SCREW_HOLE

N3
SCREW_HOLE

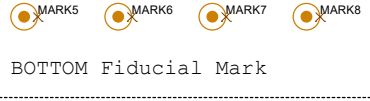
锁SODIMM的螺丝孔

锁底板的螺丝孔





TOP Fiducial Mark



BOTTOM Fiducial Mark

FuZhou X-Speed Information Technology Co.,Ltd			
Title: 22.MECHANICAL and DXF			
Size: A3	Product Model: RC01_CARRIER_V100-2026020	DESIGNER: PC	Rev: V001
Date: Wednesday, February 11, 2026	Sheet 22	of 22	